

AI Speaking Assistant

Use-Case Specification: UC5.1 Speech Recognition

Version: 1.0

Revision History

Date	Version	Description	Author
23/Jul/26	1.0	Initial RUP specification draft for UC5.1	Gia Phúc

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1. Use-Case Name

UC5.1: Speech recognition

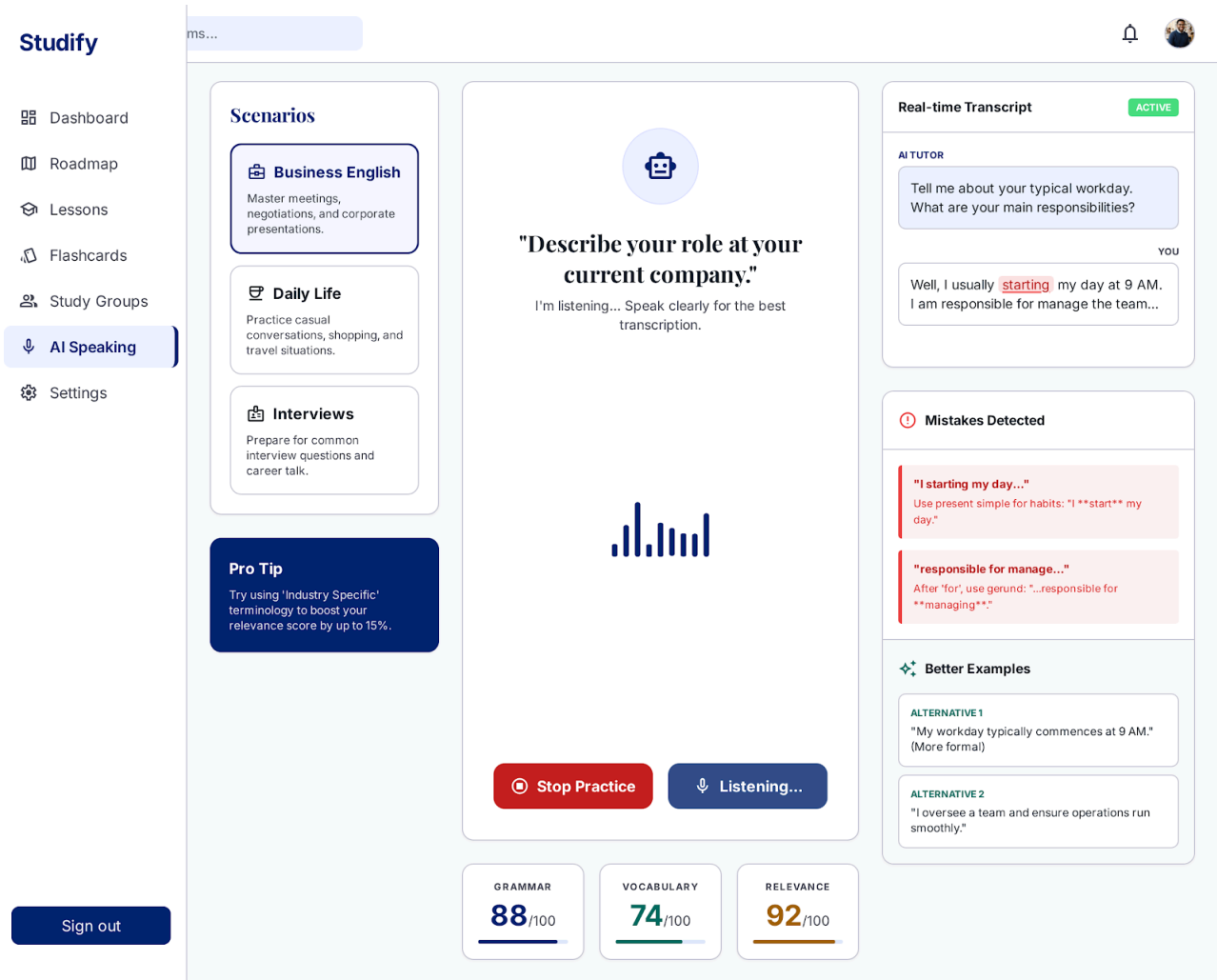
1.1 Brief Description

This use case describes how the system captures real-time spoken audio input from the Learner and processes it through the AI Engine to generate an accurate text transcription. It serves as an essential foundational service included by downstream conversation and evaluation modules.

2. Flow of Events

2.1 Basic Flow

1. The Learner begins speaking into their device microphone during an active practice turn.
2. The system streams the live audio input stream to the AI Engine's Speech-to-Text (STT) processing module.
3. The AI Engine processes the audio signal and converts it into a structured text transcript.
4. The system validates the transcribed text and returns it to the calling use case for downstream dialogue processing and evaluation.

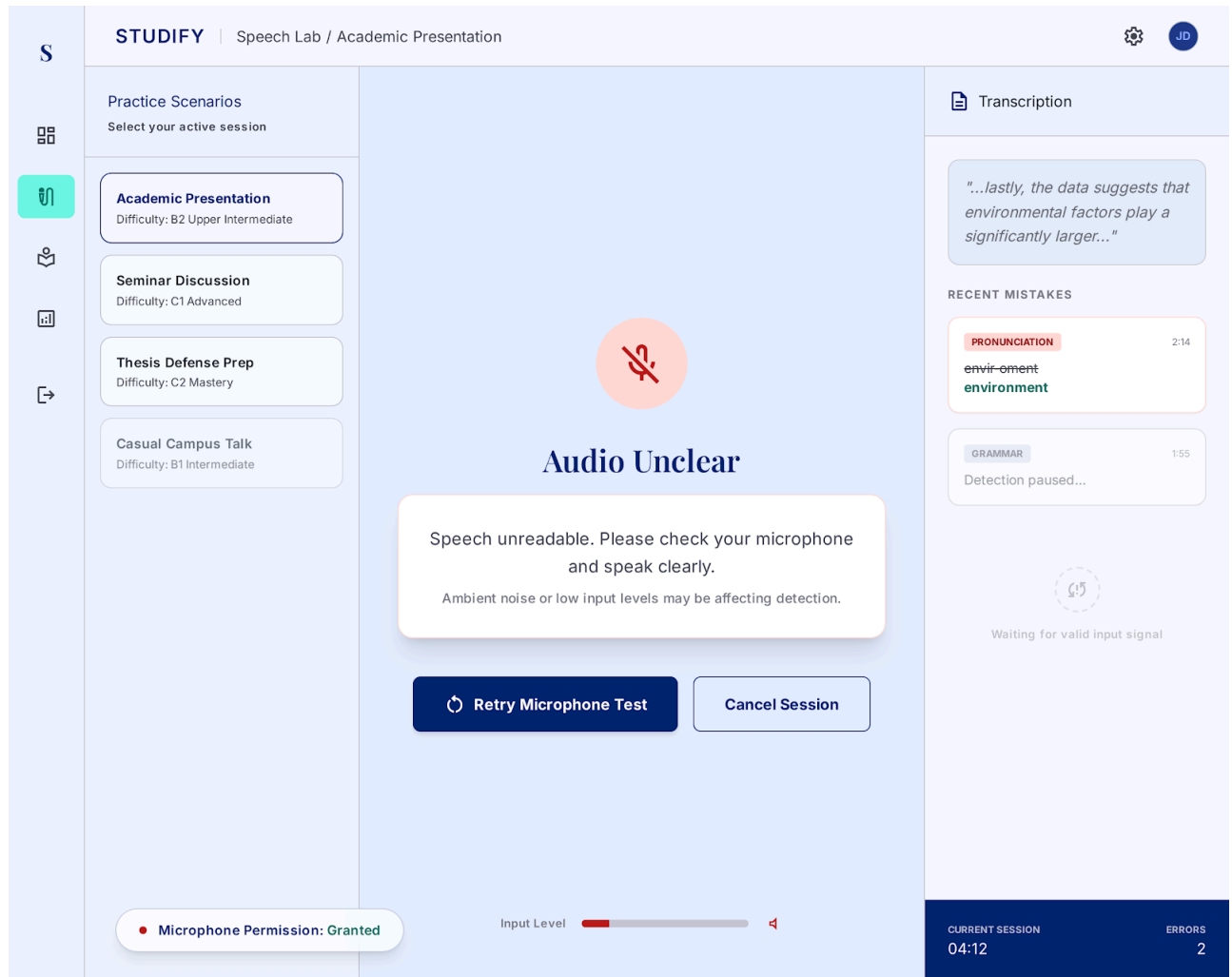


2.2 Alternative Flows

2.2.1 Audio Unclear or Low Volume

1. In step 3 of the Basic Flow, the AI Engine determines that the signal-to-noise ratio is too low or background noise obscures speech clarity.
2. The system prompts the Learner with an on-screen warning: *"Speech unreadable. Please check your microphone and speak clearly."*

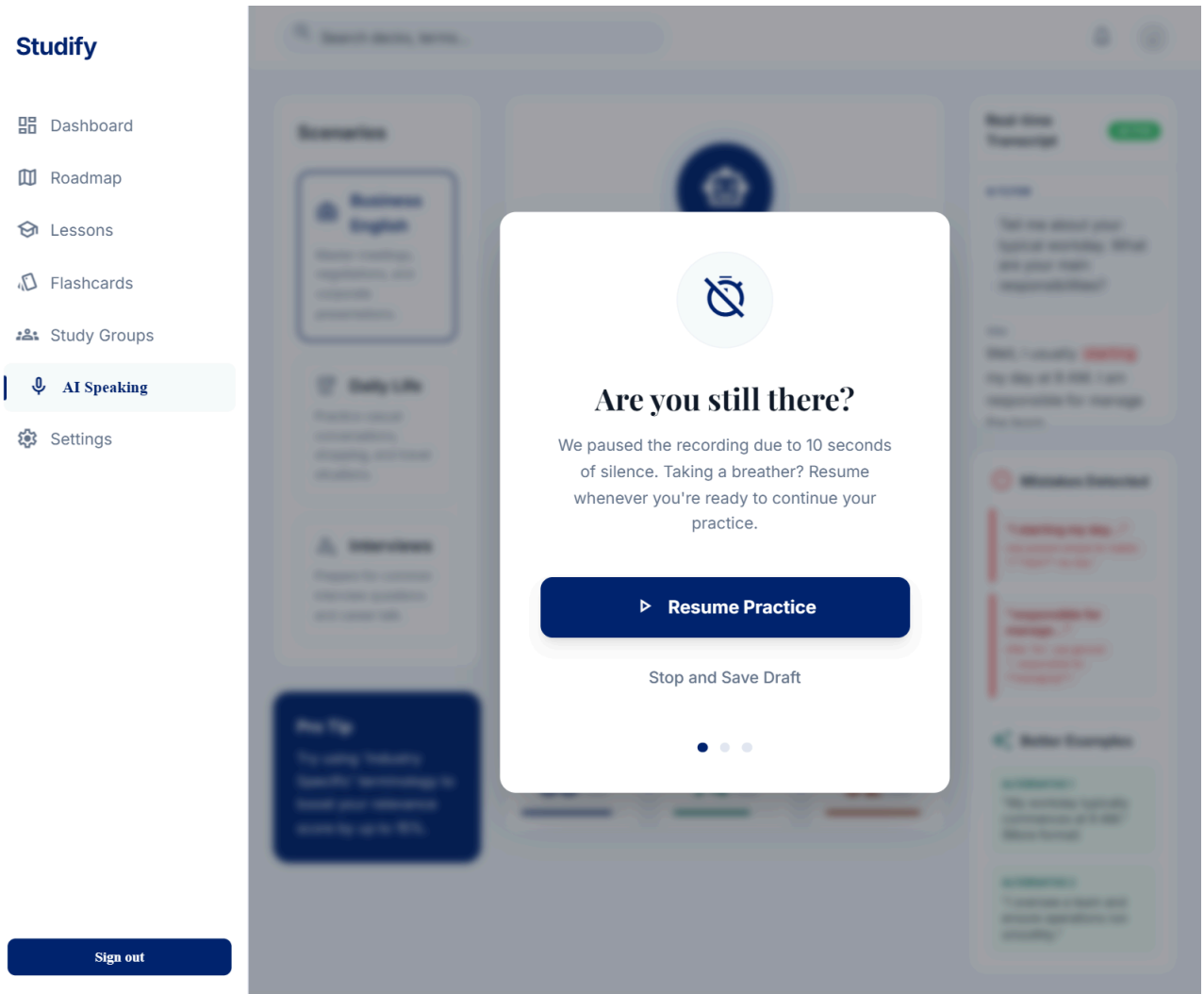
3. The Learner re-speaks their input, and the execution returns to step 2 of the Basic Flow.



2.2.2 Silence Timeout

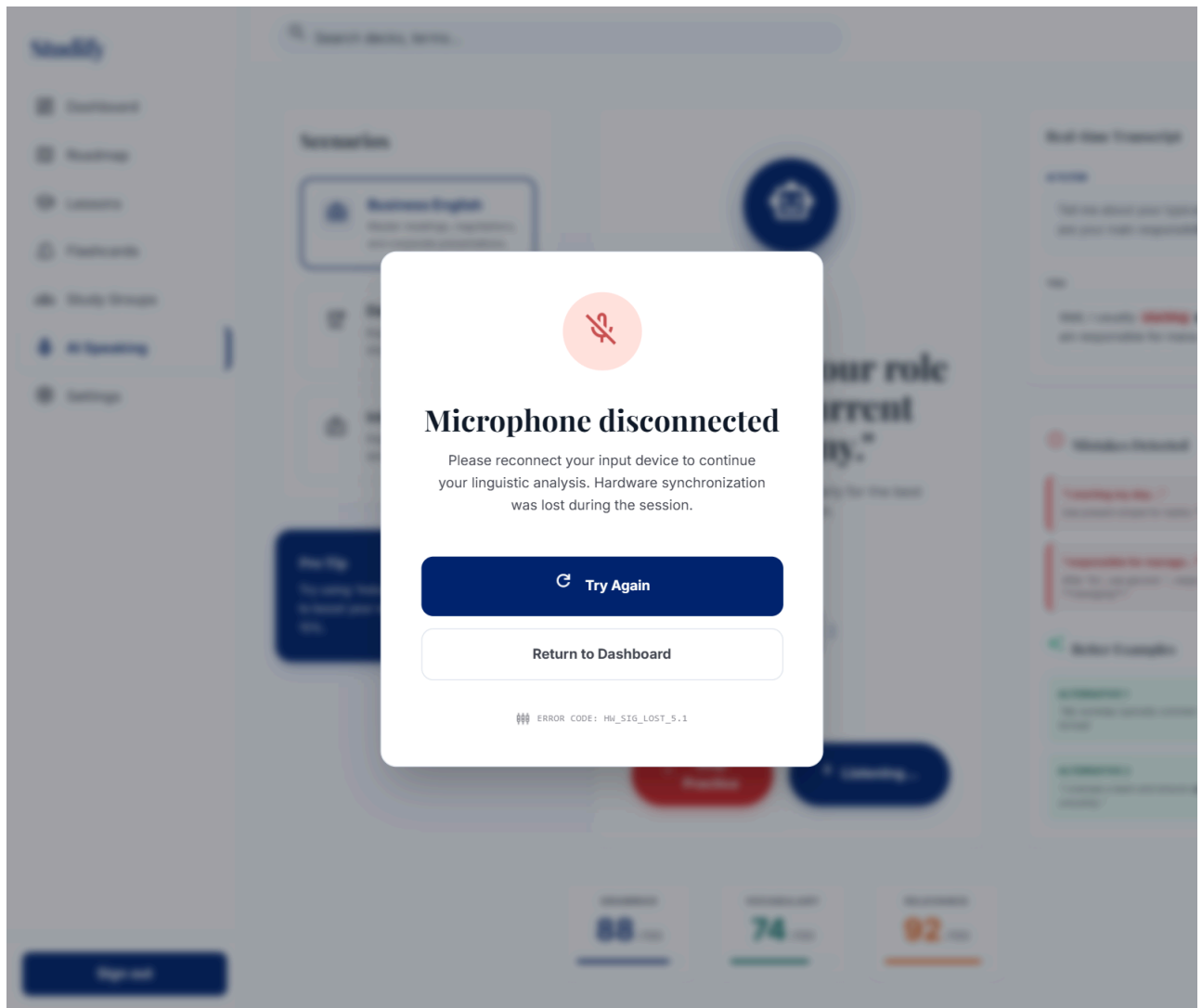
1. In step 1 of the Basic Flow, the system detects continuous silence exceeding 10 seconds.
2. The system automatically pauses active audio recording and displays a prompt asking if the user is ready.

3. Once the Learner clicks "Resume", the flow restarts at step 1 of the Basic Flow.



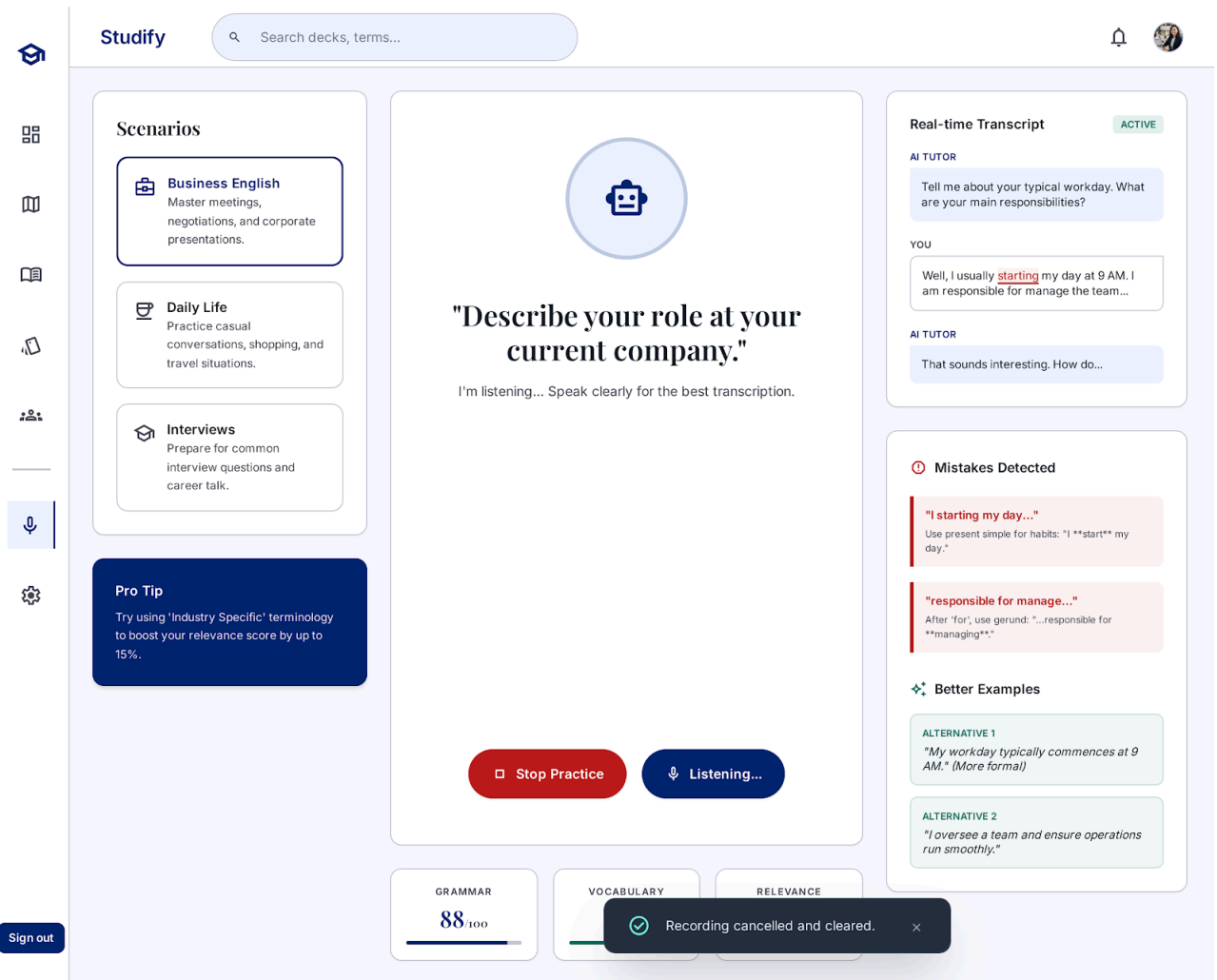
2.2.3 Audio Hardware Disconnected

1. During step 1 or step 2 of the Basic Flow, the active input microphone is unplugged or disconnected.
2. The system detects hardware loss, pauses processing, and displays an alert: *"Microphone disconnected. Please reconnect your input device."*
3. Once the system detects hardware restoration, the Learner clicks "Try Again", restarting the flow at step 1.



2.2.4 User Manually Cancels Recording

1. At any point during step 1 or step 2, the Learner clicks the "Cancel" button on the UI.
2. The system terminates audio streaming, purges temporary audio buffers, and resets the interface state back to the prompt start.



3. Special Requirements

3.1 Real-Time Processing Latency

The speech-to-text recognition pipeline must complete and return the final text transcript within 2 seconds ($< 2.0s$) following the end of user speech detection.

3.2 Multi-Language Accent Robustness

The STT model must maintain a minimum Word Error Rate (WER) accuracy of 85% or higher across varied non-native English accents.

3.3 Audio Stream Data Encryption

All live audio streamed from the client interface to the AI Engine must be encrypted in transit using TLS 1.3 / WebRTC Secure Real-Time Transport Protocol (SRTP).

4. Preconditions

4.1 Active Session State

The Learner must be authenticated and engaged in an active speaking practice session.

4.2 Microphone Permission Granted

The web browser or application permissions must explicitly allow microphone access.

4.3 Audio Input Device Detected

At least one functional audio input capture device must be active and registered by the operating system.

5. Postconditions

5.1 Audio Transcribed & Stored

The user's spoken audio is successfully transcribed into plain text and stored in the active session buffer for turn processing.

5.2 Session Metrics Updated

System logs record audio capture metadata (duration, noise level, token length) for performance analytics.

5.3 Audio Stream Closed

The active microphone capture stream is cleanly closed or set back to standby mode to save client resources.

6. Extension Points

6.1 None

This use case contains no extension points.

Use-Case Specification: UC5.2 Generate Appropriate Reply

Version: 1.1

Revision History

Date	Version	Description	Author
23/Jul/26	1.0	Initial RUP specification draft for UC5.2	Gia Phúc
6/Aug/26	1.1	Updated UC5.2 to fit with the new model	Gia Phúc

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1. Use-Case Name

UC5.2: Generate appropriate reply

1.1 Brief Description

This use case captures the end-to-end process in which the AI Speaking Assistant receives the Learner's transcribed input, evaluates the context of the active practice scenario (e.g., Business, Casual, Interview), and uses the AI Engine to construct a natural, contextually relevant response in both text and synthesized audio forms.

2. Flow of Events

2.1 Basic Flow

1. The Learner speaks their turn into the interface.
2. The system receives the verified text transcript output from UC5.1.
3. The system bundles the transcript, active scenario metadata, and prior conversation turns, submitting the payload to the AI Engine.
4. The AI Engine processes the request and generates an in-context conversational reply.
5. The system synthesizes the text response into natural-sounding voice audio.
6. The system displays the conversation text in the chat panel and plays the audio reply to the Learner.

2.2 Alternative Flows

2.2.1 Off-Topic Input Handling

1. In step 4 of the Basic Flow, the AI Engine detects that the user's input strays significantly from the scenario prompt.
2. The AI Engine generates a polite redirection response acknowledging the input while prompting the Learner back to the scenario core topic.

3. The execution resumes at step 5 of the Basic Flow.

The screenshot displays the STUDIFY AI Speaking Tutor interface. The top navigation bar includes the STUDIFY logo, "AI Speaking Tutor", a bell icon, a help icon, and a red "Stop Practice" button. A left sidebar contains icons for Dashboard, Practice, Library, Speaking, and Stats. The main content area is divided into three columns:

- Scenarios:** Three cards for "Business English" (Master meetings, negotiations, and corporate presentations), "Daily Life" (Practice casual conversations, shopping, and travel), and "Interviews" (Prepare for common interview questions and career talk).
- Pro Tip:** A blue box stating: "Try using 'Industry Specific' terminology to boost your relevance score by up to 15%."
- Listening Exercise:** A large central area with a "REDIRECTION ACTIVE" alert, a blue circle, and the text: *"I appreciate the thought, but let's try to focus on describing your professional role. What are your main responsibilities?"* Below this is a microphone icon and a "Listening..." button.
- Real-time Transcript:** A section on the right showing the user's input: "Tell me about your typical workday. What are your main responsibilities?" and the AI Tutor's response: "Well, I usually **starting** my day at 9 AM. I am responsible for manage the team...".
- Live Corrections:** A section on the right showing two errors:
 - Grammar Error:** "I **starting** my day..." with a note: "Use present simple for habits: 'I **start** my day.'"
 - Prepositional Error:** "...**responsible for manage**..." with a note: "After 'for', use gerund: '...responsible for **managing**'"
- Performance Metrics:** Three boxes at the bottom showing scores: GRAMMAR 88/100, VOCABULARY 74/100, and RELEVANCE 92/100.
- SOPHISTICATED ALTERNATIVES:** A section at the bottom right showing "Alternative 1".

2.2.2 AI Service Timeout

1. In step 4 of the Basic Flow, the connection to the AI Engine service times out (exceeds 5 seconds).
2. The system presents an error alert: *"Response generation timed out. Click to retry."*

3. The Learner clicks retry, and execution restarts at step 3 of the Basic Flow.

The screenshot displays the STUDIFY Practice Session interface. A central modal window shows a red circular icon with a white 'X' and the text: "Response generation timed out. We couldn't connect to our AI tutors. Please check your connection and try again." Below this is a blue "Retry" button. The background interface includes a sidebar with icons for Scenarios, Real-time Transcript, Mistakes Detected, and Better Examples. The Scenarios section lists "Business English", "Daily Life", and "Interviews". The Real-time Transcript section shows a conversation between an AI Tutor and a user. The Mistakes Detected section lists errors like "I starting my day..." and "responsible for manage...". The Better Examples section is partially visible. At the bottom, there are progress bars for Grammar (88/100), Vocabulary (74/100), and Relevance (92/100).

2.2.3 Inappropriate or Flagged Content

1. In step 4 of the Basic Flow, safety filters flag user input containing abusive, hateful, or explicit language.
2. The system bypasses normal conversation generation and responds with a standard safety prompt: *"Let's keep our conversation respectful and focused on the practice topic."*

3. The execution resumes at step 5 of the Basic Flow.

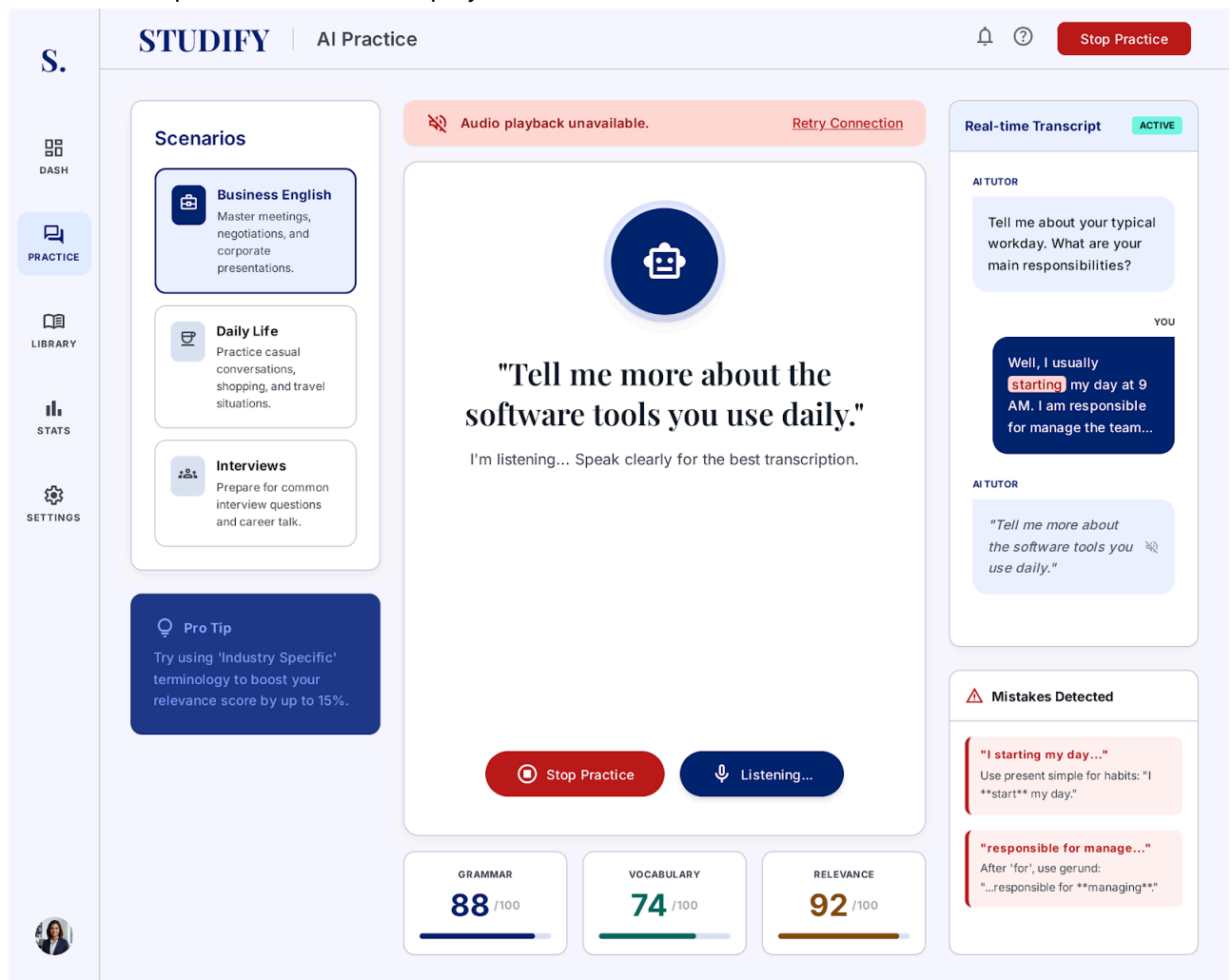
The screenshot shows the STUDIFY AI Speaking Practice interface for Session 22. The interface is divided into several sections:

- Scenarios:** Three categories are listed: **Business English** (Master meetings, negotiations, and corporate presentations), **Daily Life** (Practice casual conversations and travel situations), and **Interviews** (Prepare for common interview questions and career talk).
- Real-time Transcript:** Shows the AI TUTOR's prompt: "Tell me about your typical workday. What are your main responsibilities?" and the user's response: "Well, I usually **starting** my day at 9 AM. I am responsible for manage the team." A "Mistakes Detected" section below indicates that feedback is unavailable for flagged segments.
- Reference Phrases:** Provides two alternatives for the user's response: **ALTERNATIVE 1**: "My workday typically commences at 9 AM." (More formal) and **ALTERNATIVE 2**: "I oversee a team and ensure operations run smoothly."
- Performance Metrics:** Displays scores for Grammar (88/100), Vocabulary (74/100), and Relevance (92/100).
- Pro Tip:** Suggests using 'Industry Specific' terminology to boost the relevance score by up to 15%.
- Central Message:** A large red circle with a lock icon and the text: "Let's keep our conversation respectful and focused on the practice topic." Below this, it states: "Our AI tutor is here to help you master language in a professional and safe environment. Please refine your response to continue our lesson." and includes a "Try Another Response" button.

2.2.4 Text-to-Speech (TTS) Synthesis Failure

1. In step 5 of the Basic Flow, the TTS engine fails to synthesize audio.
2. The system displays the text reply in the chat interface and displays a non-intrusive notification: "Audio playback unavailable."

3. The flow completes without audio playback.



3. Special Requirements

3.1 Conversational Latency

The total latency for reply text generation and Text-To-Speech (TTS) rendering combined must not exceed 3 to 5 seconds to maintain natural conversation dynamics.

3.2 Natural Voice Synthesis Quality

Audio output must utilize neural TTS voices sampled at a minimum of 24kHz to ensure realistic human cadence, tone, and inflection.

3.3 Conversation Context Length

The AI Engine must maintain and process a context history window of at least 10 prior dialogue turns to ensure coherent multi-turn conversations.

4. Preconditions

4.1 Active Scenario Context

A practice scenario must be initialized, and the dialogue state must be active.

4.2 Transcribed Text Input Received

Valid text output from **UC5.1** must be passed into the dialogue handler.

4.3 AI Engine Service Availability

The system's backend AI model API must be online and responsive.

5. Postconditions

5.1 Dialogue State Updated

The newly generated AI turn (text and audio) is saved into the session history, and the state advances to wait for the user's next turn.

5.2 Audio Response Cached

Synthesized voice audio files are temporarily cached on the client to allow instant replay by the user.

5.3 History Stack Updated

The full transcript exchange is appended to the user's session history record.

6. Extension Points

6.1 None

This use case contains no extension points.

Use-Case Specification: UC5.3 Evaluate Performance on Specific Criteria

Version: 1.1

Revision History

Date	Version	Description	Author
23/Jul/26	1.0	Initial RUP specification draft for UC5.3	Gia Phúc
6/Aug/26	1.1	Updated UC5.3 to fit with the new model	Gia Phúc

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 - 5.2 UI Performance Panel Updated
 - 5.3 Analytics Trend Updated
- 6. Extension Points
 - 6.1 Performance Weakness Detected

1. Use-Case Name

UC5.3: Evaluate performance on specific criteria

1.1 Brief Description

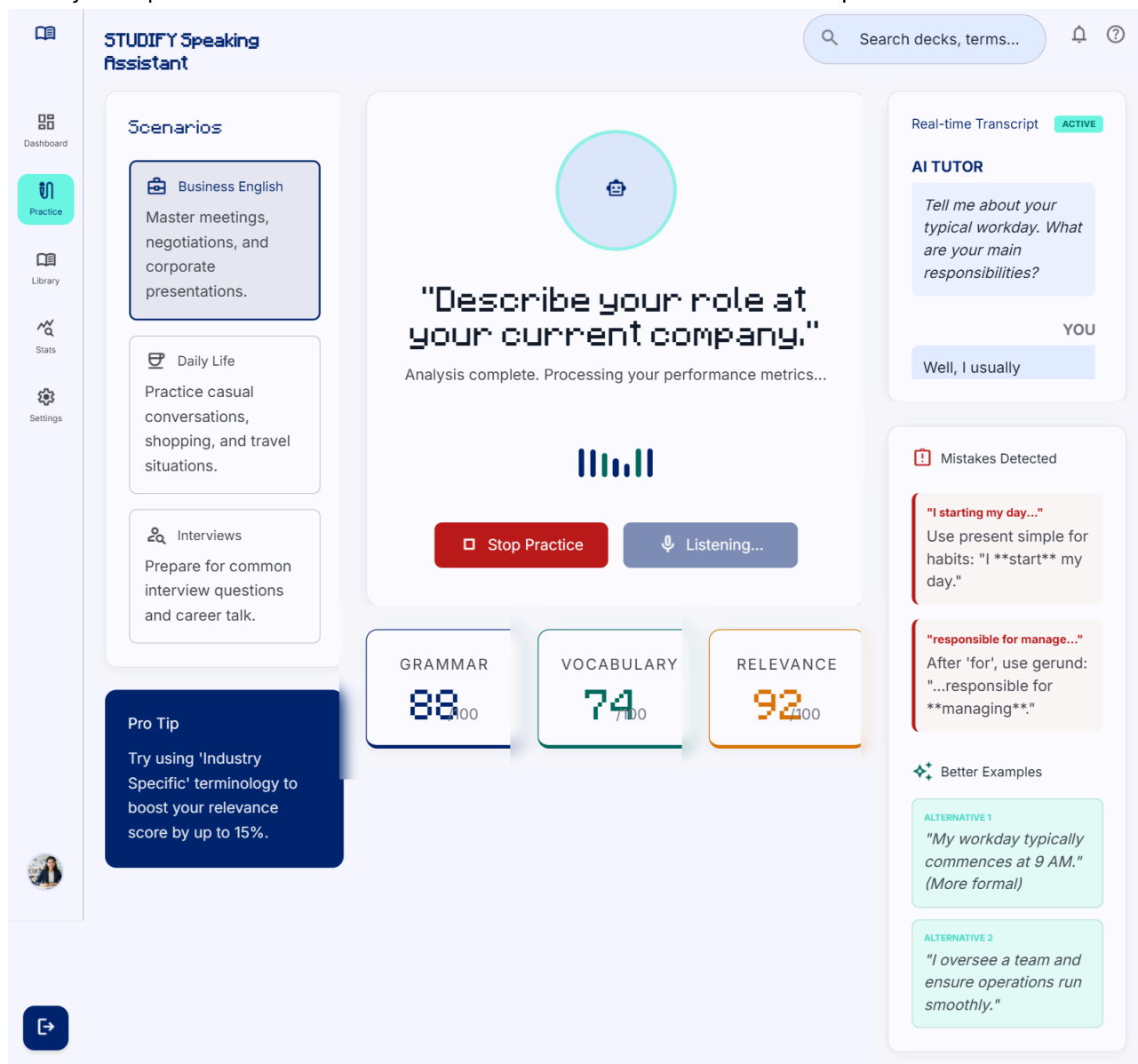
This use case details how the system analyzes the Learner's transcribed spoken speech against specific core linguistic criteria (Grammar, Vocabulary & Word Choice, and Context Relevance) to produce structured analytical scores and diagnostic feedback.

2. Flow of Events

2.1 Basic Flow

1. The system initiates a performance evaluation routine upon turn completion.
2. The system ingests the text transcript generated by UC5.1.
3. The system submits the transcript and active target evaluation rubric to the AI Engine.
4. The AI Engine evaluates performance across three strict criteria:
 - **Grammar & Syntax:** Identifies structural, tense, and agreement errors.
 - **Vocabulary & Word Choice:** Assesses lexical variety, appropriateness, and sophistication.
 - **Context Relevance:** Evaluates how accurately the utterance addresses the scenario prompt.

5. The system records numerical scores and qualitative diagnostic notes.
6. The system presents the detailed breakdown on the interface evaluation panel.



2.2 Alternative Flows

2.2.1 Brief Input Evaluation

1. In step 4 of the Basic Flow, the AI Engine determines that the user utterance is too brief (e.g., single-word responses like "Yes" or "Okay") to perform full grammatical analysis.
2. The AI Engine assigns neutral baseline scores and notes: *"Utterance too brief for detailed grammatical breakdown."*

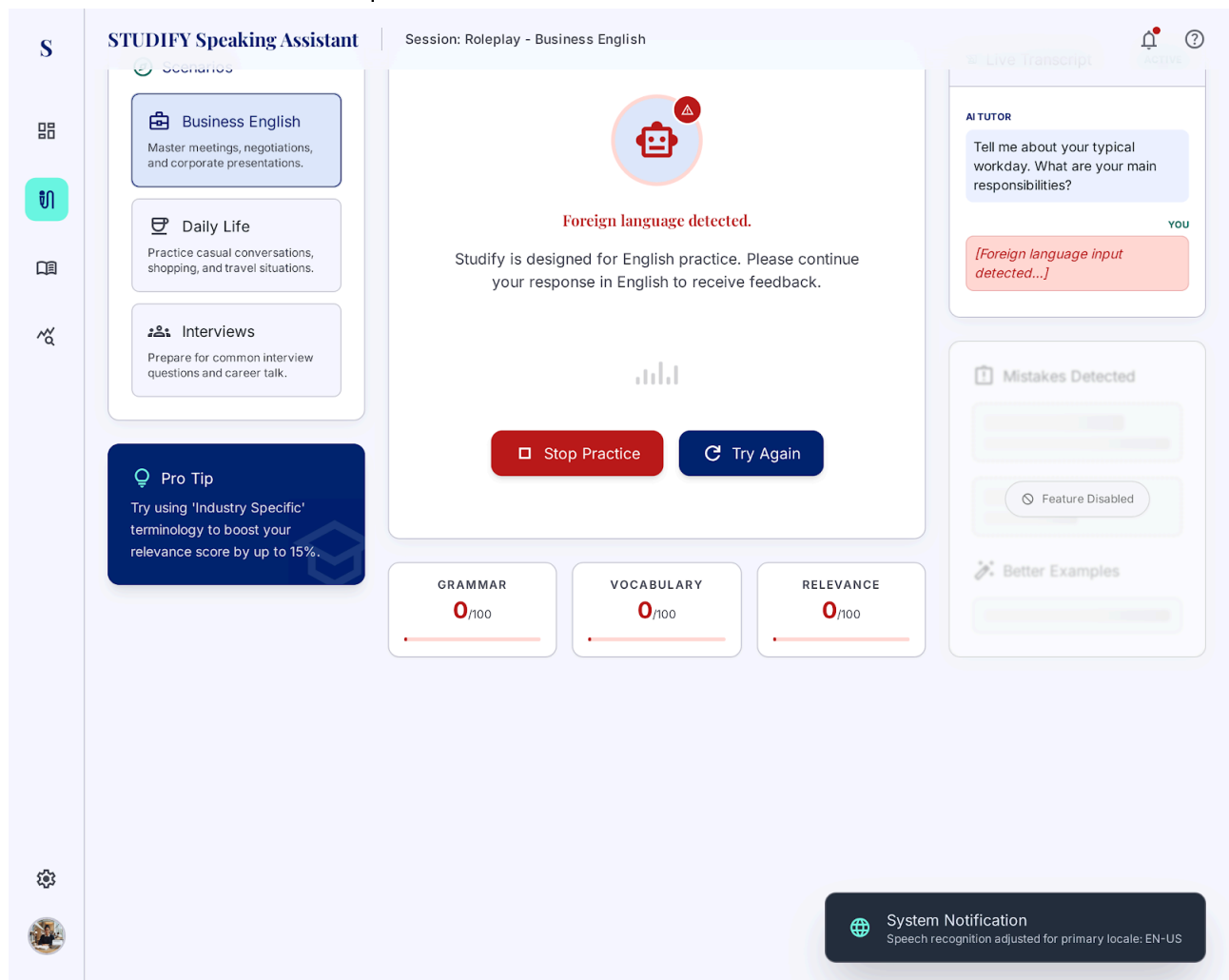
3. The execution resumes at step 5 of the Basic Flow.

The screenshot displays the STUDIFY Speaking Assistant interface. On the left, a sidebar contains icons for a home screen, a list of scenarios, a book, and a search icon. The main content area is titled "STUDIFY Speaking Assistant" and features a search bar at the top right. Below the search bar, there are three scenario cards: "Business English" (Master meetings, negotiations, and corporate presentations), "Daily Life" (Practice casual conversations, shopping, and travel situations), and "Interviews" (Prepare for common interview questions and career talk). A "Pro Tip" box suggests using 'Industry Specific' terminology to boost relevance scores. The central area shows a listening exercise prompt: "Describe your role at your current company." with a microphone icon and a status "I'm listening... Speak clearly for the best transcription." Below this, a feedback box indicates "Utterance too brief for detailed grammatical breakdown." At the bottom of the central area are "Stop Practice" and "Listening..." buttons. On the right, a "Real-time Transcript" section shows the AI tutor's prompt and the user's response "Yes." Below this, a "Mistakes Detected" section shows "N/A" with a message: "No mistakes detected in brief input. Continue speaking to get feedback." At the bottom right, a "BETTER EXAMPLES" section provides a more detailed response: "Yes, I'd be happy to. At my current job, I focus primarily on..." At the bottom of the interface, three progress bars show scores for GRAMMAR (50/100), VOCABULARY (50/100), and RELEVANCE (50/100).

2.2.2 Non-English Speech Input Detected

1. In step 4 of the Basic Flow, the AI Engine detects that the transcribed text is in a language other than English.
2. The system assigns a score of 0 for Vocabulary and Grammar, flagging the turn with: *"Foreign language detected. Please practice in English."*

3. The execution resumes at step 5 of the Basic Flow.



2.2.3 Rubric Configuration Missing

1. In step 3 of the Basic Flow, the system fails to load target scenario rubric configuration parameters.
2. The system falls back to a standardized default evaluation rubric and logs a system warning.

3. The execution resumes at step 4 of the Basic Flow.

The screenshot displays the STUDIFY Speaking Assistant interface during a practice session. The top navigation bar includes a home icon, the title "STUDIFY Speaking Assistant", a "Standard Assessment Mode" indicator, and a bell icon. The left sidebar contains icons for Home, Speak, Library, Stats, and Config. The main content area is divided into several sections:

- Scenarios:** A list of practice scenarios including "Business English" (Master business meetings, negotiations, and corporate presentations), "Daily Life" (Practice casual conversations, shopping, and travel situations), "Interviews" (Prepare for common interview questions and career talk), and "Pro Tip" (Try using 'Industry Specific' terminology to boost your relevance score by up to 15%).
- Practice Area:** A central area with a large blue circle and the prompt: "Describe your role at your current company." Below this, it says "I'm listening... Speak clearly for the best transcription." At the bottom of this section are two buttons: "Stop Practice" and "Listening...".
- REAL-TIME TRANSCRIPT:** A section on the right showing the user's transcription. It includes the AI Tutor's prompt: "Tell me about your typical workday. What are your main responsibilities?" and the user's response: "Well, I usually starting my day at 9 AM. I am responsible for manage the team." Below this, the AI Tutor's feedback is shown: "That sounds interesting. How do you plan your..."
- MISTAKES DETECTED:** A section highlighting errors in the transcription. It lists two mistakes: "I starting my day..." (Use present simple for habits: "I start my day.") and "responsible for manage..." (After 'for', use gerund: "...responsible for managing...").
- BETTER EXAMPLES:** A section providing alternative phrasings for the user's input. It includes "ALTERNATIVE 1" ("My workday typically commences at 9 AM. (More formal)") and "ALTERNATIVE 2" ("I oversee a team and ensure operations run smoothly.")
- Scorecards:** At the bottom, three scorecards are displayed: GRAMMAR (88/100), VOCABULARY (74/100), and RELEVANCE (92/100).

A warning message at the top center states: "Default evaluation rubric applied. Specific scenario rubrics could not be loaded."

3. Special Requirements

3.1 Multi-Criteria Scoring Consistency

The evaluation module must generate standardized scores (0–100 scale) along with human-readable diagnostic rationales for Grammar, Vocabulary, and Context Relevance.

3.2 Evaluation Latency

Evaluation calculations and metric scoring must be completed within 1.5 seconds following text input ingestion.

3.3 Deterministic Scoring Standards

Scoring guidelines must adhere to standardized CEFR (Common European Framework of Reference for Languages) proficiency benchmarks.

4. Preconditions

4.1 Transcribed Input Available

The Learner's spoken input must be successfully processed into text format.

4.2 Evaluation Rubric Loaded

Target scenario rubric criteria must be active in session memory.

4.3 Scenario Context Set

The specific practice domain (e.g., Job Interview, Restaurant Order) must be active to measure contextual relevance accurately.

5. Postconditions

5.1 Metrics Logged

Detailed evaluation scores and criterion breakdowns are logged in the active session report.

5.2 UI Performance Panel Updated

Visual score cards and radar charts on the learner's screen are updated with new metrics.

5.3 Analytics Trend Updated

Scores are appended to the learner's long-term progress profile database.

6. Extension Points

6.1 Performance Weakness Detected

- **Location:** Step 5 of UC5.3 Basic Flow.
 - **Condition:** If performance scores fall below preset thresholds or grammatical/vocabulary errors are flagged.
 - **Extended Use case:** UC5.4: Provide guidance on how to improve
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Use-Case Specification: UC5.4 Provide Guidance on How to Improve

Version: 1.1

Revision History

Date	Version	Description	Author
23/Jul/26	1.0	Initial RUP specification draft for UC5.4	Gia Phúc
6/Aug/26	1.1	Updated UC5.4 to fit with the new model	Gia Phúc

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1. Use-Case Name

UC5.4: Provide guidance on how to improve

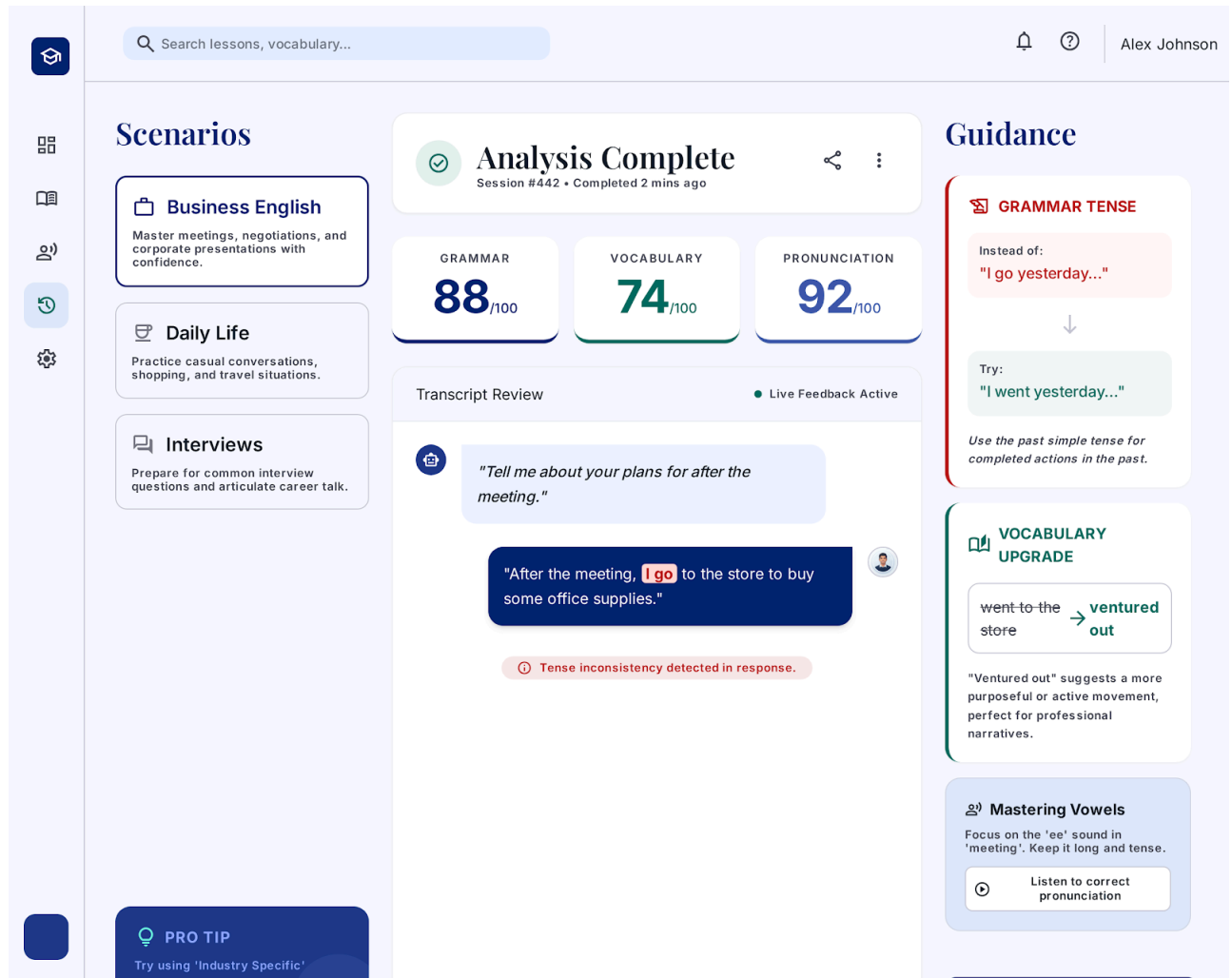
1.1 Brief Description

This use case extends **UC5.3: Evaluate performance on specific criteria** by analyzing flagged performance errors and generating personalized, actionable guidance—including corrected rephrasings and advanced vocabulary alternatives—to help the Learner rapidly improve their speaking skills.

2. Flow of Events

2.1 Basic Flow

1. Confirm that UC5.4 is triggered via the Performance Weakness Detected extension point in UC5.3 rather than executing automatically.
2. The AI Engine receives the specific error logs and lower-scoring metrics flagged during evaluation.
3. The AI Engine constructs tailored improvement guidance, including:
 - Specific sentence corrections with highlighted revisions.
 - Alternative vocabulary recommendations tailored to the context.
 - Actionable tips for improving fluency, tone, and sentence structure.
4. The system displays the personalized guidance recommendations on the feedback panel alongside the evaluation metrics.



2.2 Alternative Flows

2.2.1 Exceptional Performance Guidance

1. In step 2, if no grammatical or contextual errors were detected by **UC5.3** (flawless response):
2. The AI Engine skips corrective suggestions and instead generates high-level native speaker variations and idiomatic expressions to further elevate the user's proficiency.

3. The system displays these enhancement tips, and the flow completes.

The screenshot displays the 'UDI' interface with a 'Mastery Achieved' notification. The user is 'Alex Scholarium' at 'Mastery Level'. The interface shows three performance metrics: Grammar (100), Lexicon (94), and Nuance (98). A 'Selected Best Response' is shown, along with 'NATIVE SPEAKER VARIATION' and 'SCHOLARLY ALTERNATIVES' for improvement. The current performance estimate is 'CEFR C1 / C2'.

UDI Search sessions, terms... Alex Scholarium Mastery Level

Mastery Achieved

Flawless execution. No grammatical or syntactical errors detected in your professional self-introduction. You exhibited exceptional control over complex linguistic structures.

Metric	Score
Grammar	100
Lexicon	94
Nuance	98

Selected Best Response

"Currently, I serve as a Senior Strategist, where my primary mandate involves synthesizing complex market data to drive cross-functional alignment. I thrive in environments that necessitate both analytical rigor and creative problem-solving."

Professional Context High Lexical Density Listen to your playback

Current Performance Estimate: **CEFR C1 / C2** Export Report Continue Journey

Level Up

NATIVE SPEAKER VARIATION

More idiomatic:
"I head up the strategy department, essentially bridging the gap between raw data and actionable insights."

More fluid:
"My day-to-state revolves around translating market shifts into cohesive growth roadmaps."

SCHOLARLY ALTERNATIVES

Instead of:	Try:
"I use"	Leverage
"Think"	Conceptualize
"Change"	Transform
"Important"	Pivotal

Emotional Resonance

Your vocal projection was steady and authoritative. To reach C2 level, try introducing subtle intonation shifts when discussing "creative problem-solving" to signal passion.

2.2.2 Repetitive Error Detected

1. In step 2, the AI Engine compares flagged errors against the user's session history and identifies a recurring error pattern (e.g., repeating past tense errors 3 times in one session).
2. The AI Engine prioritizes this specific error type and attaches a dedicated mini-grammar explanation card.

3. The execution resumes at step 4 of the Basic Flow.

The screenshot displays a user interface for an English learning application. At the top, there is a search bar and a user profile for Alex Johnson. The main content area is divided into several sections:

- Scenarios:** Three cards are visible: "Business English" (Master meetings, negotiations, and corporate presentations), "Daily Life" (Practice casual conversations, shopping, and travel situations), and "Interviews" (Prepare for common interview questions and career talk).
- PRO TIP:** A blue box suggests using "Industry Specific" terminology to boost relevance scores by up to 15%.
- Practice Prompt:** A large central card with a blue circle and the text: "Describe your role at your current company." Below this, it says "the best transcription." and features a bar chart icon.
- Buttons:** Two buttons are present: "Stop Practice" (red) and "Listening..." (blue).
- Performance Metrics:** Three boxes show scores: Grammar (88/100), Vocabulary (74/100), and Relevance (92/100).
- Feedback Panel (Right Side):**
 - RECURRING PATTERN DETECTED:** 3 MISTAKES IN 1 SESSION. Topic: Past Tense Irregular Verbs. Quick Fix: Instead of "I-thinked", always use "I thought". Quick Drill: 2 min.
 - Real-time Transcript:** Labeled as ACTIVE. It shows a conversation between an AI Tutor and the user. The AI Tutor asks: "Tell me about your typical workday. What are your main responsibilities?". The user responds: "Well, I usually starting my day at 9 AM. I am responsible for manage the team...". The AI Tutor responds: "That sounds interesting. How do you handle the morning catch-ups?".
 - Better Alternatives:**
 - MORE FORMAL:** "My workday typically commences at 9 AM."
 - MORE NATURAL:** "I oversee a team and ensure operations run smoothly."

2.2.3 Learner Dismisses Guidance Cards

1. During step 4 of the Basic Flow, the Learner clicks "Hide Feedback" or "Skip Guidance".
2. The system minimizes the feedback panel, keeping only the high-level numerical score visible, and logs the user preference.

The interface features a top navigation bar with a search bar labeled 'Search decks, terms...', a notification bell, a help icon, and the user's name 'Alex Rivera' with the title 'Advanced Learner'.

Scenarios

- Business English**: Master meetings, negotiations, and corporate presentations.
- Daily Life**: Practice casual conversations, shopping, and travel situations.
- Interviews**: Prepare for common interview questions and career talk.

Pro Tip: Try using 'Industry Specific' terminology to boost your relevance score by up to 15%.

Central Practice Area

"Describe your role at your current company."

I'm listening... Speak clearly for the best transcription.

Buttons: Stop Practice, Listening...

Performance Metrics

GRAMMAR	VOCABULARY	RELEVANCE
88 /100	74 /100	92 /100

Transcript Panel (ACTIVE)

AI TUTOR: Tell me about your typical workday. What are your main responsibilities?

YOU: Well, I usually starting my day at 9 AM. I am responsible for manage the team...

AI TUTOR: That sounds interesting. How do you handle conflict in your team?

FEEDBACK DETAILS >

3. Special Requirements

3.1 Actionable Feedback Formatting

Guidance output must strictly provide clear "Before / After" rephrasing examples (e.g., "*Instead of: 'I go yesterday', try: 'I went yesterday'*").

3.2 Pedagogical Tone

All corrective feedback must maintain an encouraging, constructive, and supportive tone.

3.3 Visual Contrast and Readability

Grammar corrections and vocabulary replacements must be highlighted using clear, high-contrast color coding (e.g., red strike-throughs for errors, green text for corrections).

4. Preconditions

4.1 Evaluation Completed

UC5.3: Evaluate performance on specific criteria must have executed and generated diagnostic outputs.

4.2 Weakness or Error Flagged

Specific areas for improvement or errors must be identified in the step outputs of **UC5.3**.

4.3 Learner Profile History Loaded

Prior session performance logs must be accessible to detect repeated mistake patterns.

5. Postconditions

5.1 Recommendations Saved

Personalized learning suggestions and rephrasing cards are appended to the user's history profile for future review.

5.2 Practice Deck Updated

Flagged grammar points and vocabulary items are automatically saved to the user's personal review list.

5.3 Feedback Display Rendered

The guidance cards are rendered on the user interface feedback panel.

6. Extension Points

6.1 None

This use case contains no extension points.